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Recognition Of Valid QR Codes with Machine Learning

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***ABSTRACT*- The widespread use of Quick Response (QR) codes has made QR codes an attractive target for cyberattacks, posing a security and privacy concern. Quick Response (QR) codes have revolutionized marketing strategies by providing a fast and efficient way to connect offline and online platforms. However, the security issues arise as the usage of QR codes. This paper introduces an automatic detector that identifies QR code images and lets the user know whether it is malicious or benign. To achieve this, a dataset of 10,000 original and malicious QR code images (both) is taken. Convolutional Neural Network (CNN) is employed to classify original and malicious codes. Additionally, histogram density analysis is incorporated to enhance feature identification, leading to improved classification accuracy. This paper implemented ResNet50, MobilenetV3 and InceptionV3 CNN models for evaluating the abilities to predict the malicious codes. The InceptionV3 model emerges as the most promising, attaining an accuracy of 98.13%, offering an efficient and accurate solution for document forensics and counterfeit detection.**

Keywords- convolutional neural networks (CNN), InceptionV3, ResNet50, MobileNetV3 histogram density analysis, malicious codes, benign codes, machine learning.

1. INTRODUCTION

Quick Response (QR) codes have become ubiquitous in today's digital age, being used in a wide range of applications, from marketing and advertising to logistics and inventory management. Denso Wave developed QR codes in 1994; they can be used for payments, advertising, tracking your health, and more. However, the ease of creating and distributing QR codes has also made them vulnerable to fraudulent activities, such as phishing and malware attacks. Therefore, it is essential to develop a reliable and efficient method for recognizing valid QR codes and distinguishing them from malicious ones. Phishing attacks that use malicious QR codes can compromise sensitive data. This study focuses on the differentiation between the original and malicious codes.

The function of two-dimensional barcodes have advantages over conventional one-dimensional barcodes since they can contain more data and are more adaptable. Smartphones can quickly read QR codes, enabling users to connect to websites, view digital material, and activate features like email and SMS. They are machine-readable; hence their content cannot be understood by humans. With examples of its use in many nations, QR codes have found use in marketing, product information, business cards, and more.

Validation of QR codes is crucial to ensure accuracy and reliability. It confirms data integrity, assess error correction, verify link security, and comply with standards. QR codes are two-dimensional barcodes widely used for encoding data such as URLs, text, or other information. Validating them ensures they serve as reliable conduits for accessing supplementary information and enhancing the overall user experience. They do not let the user fall into the prey's hand and let them be attacked.

Convolutional Neural Networks (CNNs) are among the machine learning techniques that have been developed to classify and eliminate different kinds of codes. The study presents a dataset of 10,000 original and malicious QR codes for training with deep learning models particularly in CNN architecture, like MobilenetV3, ResNet50 and InceptionV3 , in order to improve the reading of QR codes. To tackle noise in QR codes of various sizes, the research uses a hybrid approach that combines histogram analysis and machine learning.

MobilenetV3 focuses on retaining high accuracy while lowering computing costs. It is appropriate for environments with limited resources because of its simplified design and optimisation for jobs like image classification and object recognition. ResNet50 is an efficient deep learning architecture used for image identification. Its main development is residual learning, which circumvents the vanishing gradient problem and allows for the construction of incredibly deep neural networks. It is a well-liked option for computer vision applications such as object detection, picture classification, and others due to its exceptional performance and efficiency.

Convolutional neural network (CNN) architecture InceptionV3 is intended for object detection and image classification. It has sophisticated modules for effective information processing, such as inception blocks. Because of its strong feature extraction capabilities and excellent

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accuracy, InceptionV3 is frequently used in computer vision applications such as image recognition, scene interpretation, and medical image analysis. The paper aims to create a QR code scanning solution of differentiating between benign and malicious codes that can be used in real time applications, but is subject to constraints related to scope, time, cost, quality, resources, risks, and integration. Deep learning models were used since they provided with better accuracies in previous researches than machine learning algorithms.

The remainder of this paper is divided into four sections: Section II reviews the previously undertaken research work. Section III contains details regarding the dataset and the models utilized in the research. Section IV comparatively evaluated the performance of the implemented models. Section V contains the concluding remarks of the research

II. LITERATURE REVIEW

TABLE I. A SUMMARY OF LITERATURE REVIEW

Title	Year	Dataset	Model/Technique Used	Key Findings
Secure Real-Time Artificial Intelligence System against Malicious QR Code Links [1]	2021	Dataset of 100,000 benign and malicious URLs	Naive Bayes, Support Vector Machine, Logistic Regression, K-Nearest Neighbors, Decision Tree.	Naïve Bayes - 73.92% SVM – 84.67% LR – 85.72% KNN - 89.61% Decision Tree - 90.24%
Performance comparison of machine learning driven approaches for classification of complex noises in quick response code images [5]	2022-2023	Dataset of 80,000 Clean and Noisy Images	Seventeen pre-trained deep learning models, Two machine learning algorithms.	Achieved an accuracy of 87.6% using deep learning models, utilizing XceptionNet and machine learning algorithms achieved 63.13% accuracy.
Implementing Deep Convolutional Neural Networks for QR Code-Based Printed Source Identification [3]	2023	42240 QR Codes, BW images	AlexNet, GoogleNet, DenseNet201, MobileNetv2, ResNet18, ResNet50, VGG16, Customized CNN model	AlexNet - 9.09 % GoogleNet - 89.8% DenseNet20 - 97.8 % MobileNetv2 – 99.3% ResNet18 – 99.4% ResNet50 – 99.5% VGG16 – 9.09% Customized CNN model – 99.7%
Cyber attack detection with QR code images using lightweight deep learning models [6]	2022	6000 QR code images	MobileNetV2 ShuffleNet	MobileNetV2 – 90.28% ShuffleNet – 88.78%
Detecting Malicious URLs using Machine Learning Techniques [4]	2022	20,000 URLs	Random Forest Multi layer perception KNN SVM NB	Random Forest – 97.69% Multi layer perception– 97.28% KNN – 96.25% SVM – 96.10% NB – 93.86%

In this section, we provide a brief overview of recent advancements in Valid QR Code detection through the application of Machine learning and Deep learning techniques and look into their work.

One notable study by Zahrani 2021 et al. [1] conducted a secure real-time computational intelligence system against valid QR code links, having a dataset of 1,00,000 benign and malicious QR code URLs; the authors applied types of AI classifiers. The authors reported Decision Tree classifier achieving the highest overall accuracy of 90.24%. However, the others classifier such as SVM and Naïve Bayes were unable to predict the validity of the links accurately.

Another study by Ahmad Bilal Wardak 2023 et al. [5] discuss the problem of noise presence in QR code images and propose CNN model-based system for classification. They use various deep learning models, including InceptionV3 and XceptionNet whose architecture incorporated SoftMax output layer, ReLU layer to classify noisy and original QR

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code images. Their experimental results show that InceptionV3 and XceptionNet models outperformed others by achieving 87.6% accuracy each. The authors also used machine learning algorithms such as NB and Decision Tree which gave the maximum accuracy of 63.13%. These scores indicated the importance of incorporating additional convolutional layers for tacking more complex images.

In a study by Ahmad S Rafsanjani 2022 et al. [2] they talk about threats to the privacy of android and provide guidelines for a usable, secure application designing. And in another prior study by Min-Jen Tsai 2023 et al. [3] propose a customized CNN model that demonstrates better results in identifying printed sources of grayscale and color QR codes with less computational power and training time. Various deep learning models, specifically AlexNet, GoogleNet, and ResNet, were employed. These models were subjected to experimentation using three different optimizers on separate black and white (BW) and color QR code datasets. Remarkably, the customized CNN model exhibited the highest level of accuracy of 99.7%, followed closely by ResNet50 model attaining accuracy of 99.5%. They suggests that machine learning, particularly deep learning techniques such as CNNs, can be effectively used for recognizing valid QR codes. Subsequently, another study by Malak Aljabri et al. [4] a review paper discuss that Support Vector Machines (SVM), Random Forest (RF), and Naive Bayes (NB) being frequently used, Convolutional Neural Networks (CNN) and XGBoost emerged as high-performing alternatives along with SVM, achieving an impressive accuracy of 99% over the classifiers which obtained the accuracy of 84.74% by the Bayesian network classifier.

In a study by Frank Vanhoenshoven 2023 [14], the cybersecurity community conducted an empirical investigation using blacklisting services, which are created by techniques including manual reporting, honeypots, and web crawlers using heuristic site analysis, to discover harmful websites. The work evaluates classifiers (Naïve Bayes, SVM, MLP, Decision Trees, Random Forest, k-NN) on a dataset of 20,000 URLs, treating malicious URL detection as a binary classification problem. According to simulations, most techniques produce predictions that are accurate enough, with Random Forest and Multi-Layer Perceptron showing a remarkable of 97.28% accuracy rate.

Another study by Yusuf Alaca 2022 et al [6] emphasized on the intrusion detection of the codes by training of CNN models, MobileNetV2 and ShuffleNet along with Harris hawk Optimisation for selecting the features. By implementing the image processing techniques in machine learning methods with the images of 6000 QR codes, the CNN models proved to be successful in detecting the attacks with 95.89% accuracy.

QR codes have gained popularity for instant content access due to their high information density and error correction. Yet, they've become a viable attack vector, particularly for phishing. Malicious codes are distributed deceptively, often placed over legitimate QR codes. Users rely on reader software to discern benign from malicious QR codes. [7] Instead of checking the URL encoded in a QR code against appropriate threat databases, a number of so-called secure QR code scanner apps just show the user the URL encoded in a QR code. Furthermore, a lot of apps lack support for fundamental security measures like alerting the user to the URL and requesting confirmation before opening it in a browser. The study's most concerning finding is that only two of the apps under investigation validate the redirected URL linked to a QR code [10]. Additionally, we used a collection of safe, and malicious QR codes that we gathered from various sources to evaluate the pertinent. These previous research outcomes provide a solid foundation for the current study, which seeks to extend this knowledge by implementing and comparing the deep learning models. Table I gives a comprehensive summary of the literature review conducted. The insights gained from these prior studies underscore the potential of CNNs in automating and enhancing the accuracy of QR code related tasks, which is particularly relevant in the context of valid QR code recognition.

III. PRELIMINARIES AND PROPOSED METHODOLOGY

In this study, we aimed at identifying malicious and benign QR codes, the process involves the following sequential steps.

A. Preliminaries:

1) InceptionV3: In contemporary computer vision, we have a highly efficient approach known as Transfer learning. Transfer learning is a machine learning technique that capitalizes on the use of a pre- trained neural network. In this context, the neural network model we are utilizing are InceptionV3, MobileNetV3 large, and ResNet50.

InceptionV3 leverages a meticulously crafted architecture featuring inception modules. These modules, designed to capture features at multiple scales concurrently, employ parallel convolutional operations with varying filter sizes (1x1, 3x3, and 5x5). This multi-scale approach enables InceptionV3 to extract a diverse range of features, enriching its ability to represent complex visual patterns. Additionally, the model employs auxiliary classifiers at intermediate layers to facilitate training and enhance regularization.

2) ResNet50: ResNet50, another pre-trained model renowned for its remarkable performance in various image classification tasks, was our second exploration. Its architecture hinges on residual blocks, each containing skip connections that enable unimpeded information flow through the network. This design mitigates the vanishing gradient problem, allowing for deeper networks with enhanced representational capacity. To further bolster feature extraction

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and training, ResNet50 incorporates residual connections and strategically positioned auxiliary classifiers within its architecture. These classifiers, located at intermediate layers, contribute to improved gradient flow and regularization.

3) MobileNetV3 large: A pre-trained convolutional neural network designed for efficient performance on mobile and embedded devices. Its core building blocks, depth wise separable convolutions, effectively reduce computational costs while maintaining accuracy. These blocks factorize a standard convolution into two steps: a depth wise convolution applying a single filter per input channel, followed by a pointwise convolution for channel mixing. This factorization significantly reduces the model's parameter count and computation demand. To enhance feature extraction capabilities, MobileNetV3-Large also employs inverted residual blocks and squeeze-and-excitation mechanisms. We further augmented its feature extraction power by adding layers on top of the pre-trained base model.

B. Data preprocessing:

In the preprocessing phase of this study, we carefully prepared the QR code dataset to ensure optimal input for the chosen models. Eq.1 is a step involved pixel value normalization, which we achieved by dividing each pixel value by 255. This transformation effectively scales the values within the range of [0, 1], promoting consistency and facilitating faster model convergence during training. Normalization is particularly beneficial when employing gradient descent algorithms, as it improves stability and optimizes the optimization process.

datagen = ImageDataGenerator(rescale=1.0/255.0) (1)

Furthermore, we utilized data augmentation techniques to increase the size and diversity of the training dataset. These techniques, summarized in Table II, were employed using the ImageDataGenerator class from the Keras library. Data augmentation helps the model learn robust features that generalize well to unseen data, improving its overall performance and resilience to variations in real-world QR code images.

TABLE II. PREPROCESSING PARAMETERS FOR ALL CNN MODELS.

Parameter	Value
Rotation Range	40 degrees
Width Shift Range	0.2
Height Shift Range	0.2
Shear Range	0.2
Zoom Range	0.2
Horizontal Flip	True
Fill Mode	Nearest

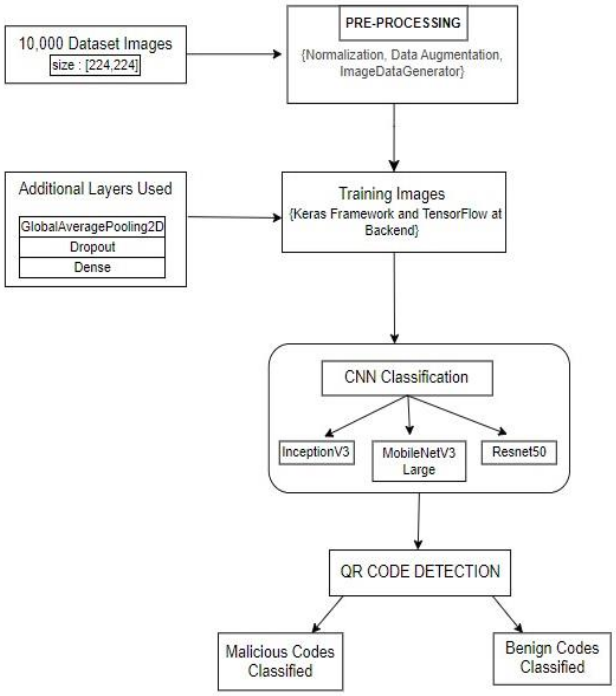

Figure 1: Methodology

IV. RESULTS AND DISCUSSIONS

In this section, the results obtained after implementing our Deep Learning models on the QR code dataset are summarized.

A. Dataset Collection:

The dataset utilized for the training and evaluation of our proposed model was sourced from Kaggle[1]. It consists of 10,000 high-resolution QR code images belonging to two classes: “Benign”, and “Malicious”.

B. Experimental Settings:

All the three models were trained for thirty epochs. The initial dataset was split and 70 % was utilized for training, and 30 % for testing. Hyperparameters such as batch size, optimizer, and preprocessing were chosen and implemented through trial and error. The batch size was selected as thirty-two, ADAM optimizer and binary cross-entropy. The models were trained using the Keras framework with TensorFlow as the backend. TensorFlow was used due to its versatility. The layers of the base model are frozen, ensuring that their weights are not updated during training to preserve pre-trained knowledge. Following the base model, a Sequential model is constructed, incorporating a global average pooling layer for further spatial reduction, facilitating a more compact representation of features. Followed by a dense layer was added with the layer having 128 units and ReLU activation

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which is introduced for additional feature extraction, the output layer with one unit and sigmoid activation for binary output. Initializing the weights with l2 regularizer. To prevent overfitting, a dropout layer was placed with a rate features of '0.6'. Dropout layers are placed after the global average pooling layer. Dropout is a regularization layer strategically placed to reduce overfitting instances.

TABLE III. HYPERPARAMETER FOR ALL CNN MODELS

Parameter	Value/Type
Image Size	224x224
Training/Testing	70/30
Optimizer	Adam
Batch Size	32
Max Epochs	30
Activation	Sigmoid/ReLU
Loss Function	Binary Crossentropy
Initial Learn Rate	10^{-4}
Dropout	0.6

C. Experimental Results and Comparative Analysis

In our comprehensive evaluation of three distinct models, InceptionV3, MobileNetV3 Large, and ResNet50, the obtained results reveal nuanced patterns across the models.

InceptionV3 exhibited a commendable ability to capture intricate hierarchical features, leading to robust classification with training accuracy of 98.13% and validation accuracy of 98.00%, The model's ability to capture intricate hierarchical features contributes to its stability during training. Figure 2 shows InceptionV3 emerges as the frontrunner with a learning curve displaying notable consistency and promising results. A comparative summary of all three models along with comparison of the accuracy and loss is shown in Table IV.

TABLE IV. PERFORMANCE OF CNN MODELS.

Model	Epochs	Training Accuracy	Training Loss	Validation Accuracy	Validation Loss
MobilenetV3	30	96.87%	12.34	90.47%	28.53
ResNet50	30	98.04%	6.98	98.30%	6.80
InceptionV3	30	98.13%	6.11	98.00%	5.17

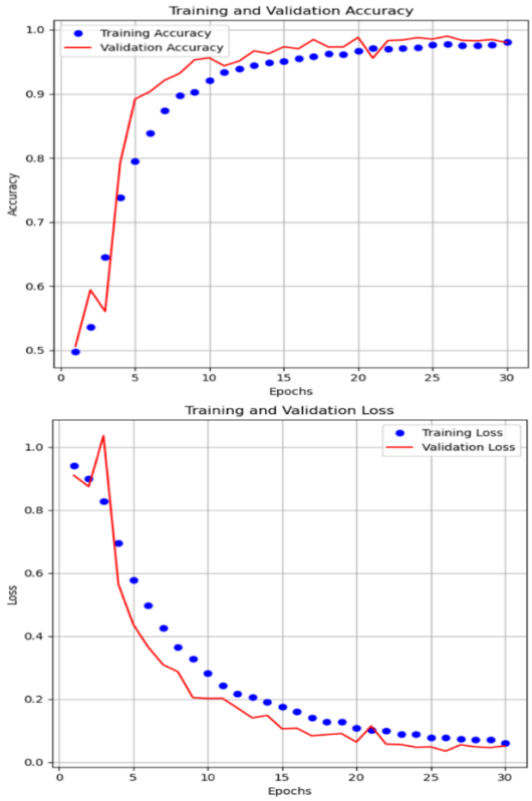

Figure 2: InceptionV3 Graph -Training and Validation Accuracy(left), Training and Validation Loss(right)

ResNet50 model showing a training accuracy of 98.04% and validation accuracy of 98.30%, with more loss than InceptionV3. However, its training process showcased some unsteadiness, necessitating careful consideration for future refinement. Figure 3 shows ResNet50 model showcase a learning curve with a slightly noticeable degree of inconsistency, reflecting the intricate balance between model complexity and performance.

MobileNetV3 Large has shown unsatisfactory performance with an accuracy of 96.87% and validation accuracy of 90.47% which is the lowest among the three models, showing a clear case of overfitting and depicting the model's inability to generalize well on unseen data. Figure 4 shows the inconsistent learning curve progress in model accuracy and loss.

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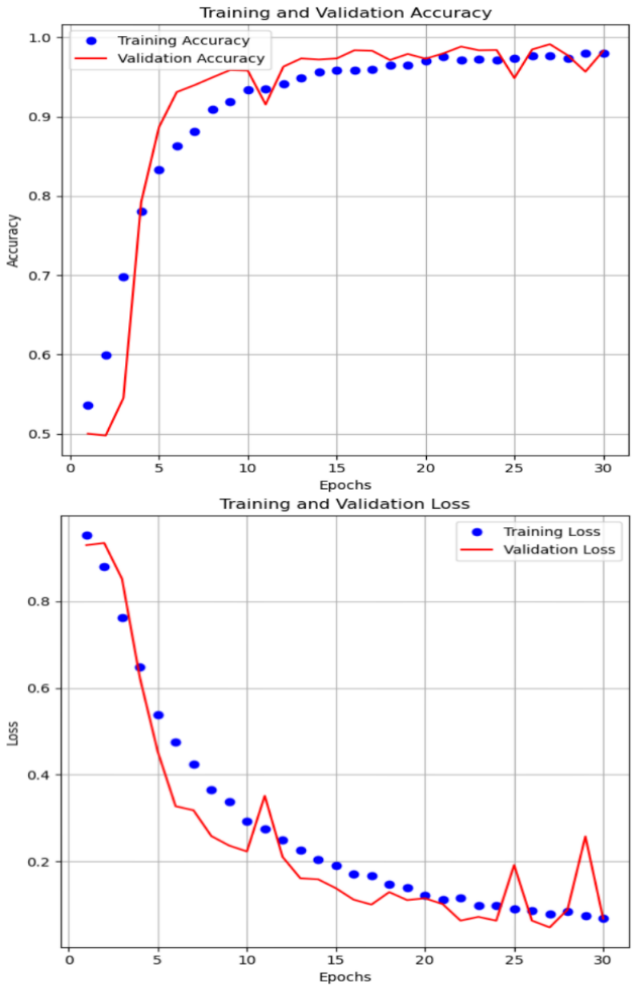

Figure 3: ResNet50 Graph -Training and Validation Accuracy(left), Training and Validation Loss(right)

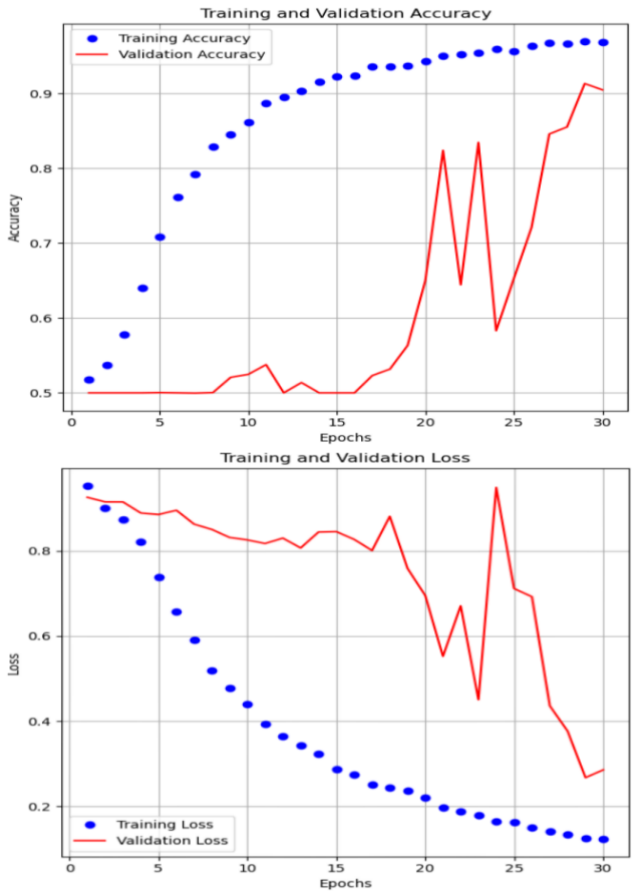

Figure 4: MobilenNetV3 large Graph -Training and Validation Accuracy(left), Training and Validation Loss(right)

Figures 5a. and 5b. shows a graph-based comparison of all the three models. The validation loss is highest in the case of MobilenNetV3 large but reasonably low in ResNet50 and minimum in the proposed model InceptionV3.

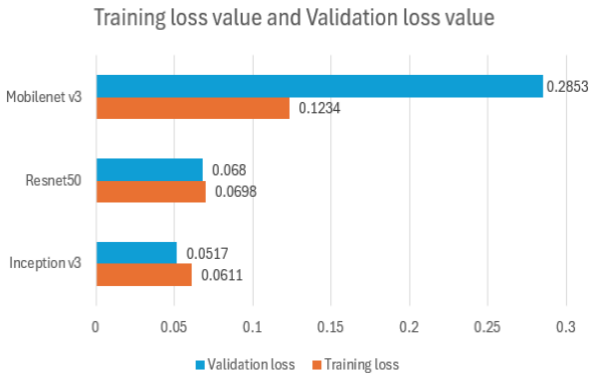

Figure 5a. Comparative Loss Analysis of three models

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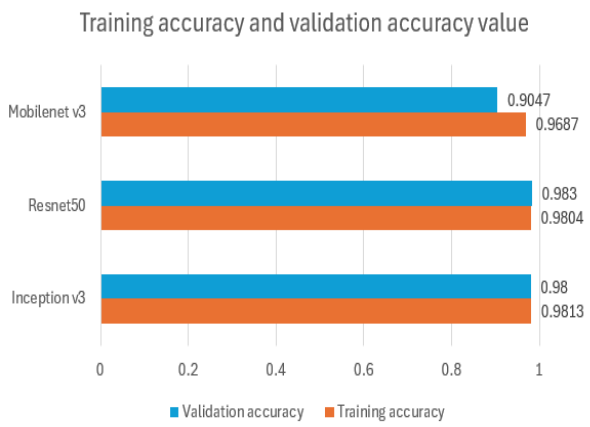

Figure 5b. Comparative Accuracy Analysis of three models

V. CONCLUSION

The comparative evaluation of implemented models revealed the remarkable effectiveness of the InceptionV3 CNN model in accurately recognizing valid QR codes. TShe proposed model exhibited exceptional performance, boasting an impressive accuracy of 98.13% on the training dataset and 98.00% on the validation dataset, while also exhibiting minimal loss compared to alternative models such as MobileNetV3 large and ResNet50. Notably, these models exhibited inconsistencies in their learning curves. Leveraging its high accuracy, the proposed model holds significant potential for real-time security applications, allowing for the identification and separation of legitimate QR codes from fraudulent ones. In an era where the reliance on QR codes for various purposes continues to grow, the utilization of machine learning and deep learning techniques for QR code recognition represents a significant advancement in security measures. These innovations offer a promising avenue for enhancing security and safeguarding users against potential threats. Through these technological advancements, our digital interactions can be fortified, ensuring that QR codes remain a reliable and secure tool in our modern world.

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